Sumitabh Ahirwar

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Education

- **Motilal Nehru National Institute of Technology Allahabad** *2023-2027*
  Bachelor of Technology in Computer Science and Engineering CPI : 8.30 (Till VIth sem)
- **Govt. Excellence Higher Secondary School** *2022*
  12th, Board of Secondary Education, Madhya Pradesh **Percentage:** 91.2
- **Govt. Excellence Higher Secondary School** *2020*
  10th, Board of Secondary Education, Madhya Pradesh **Percentage:** 96.7

Projects

•Researchos | AI Research Report Generator

- Tech-stack: Python, FastAPI, LangChain, Mistral AI, and REST APIs for web, PDF, RAG and hybrid research workflows.
- Implemented a multi-agent pipeline with Query Planner, Web Search, PDF Retriever, Analyzer, Report Writer, Citation Checker, and Critic agents for structured report generation.
- Developed PDF RAG using PyPDFLoader, Mistral embeddings, ChromaDB vector storage, chunking, semantic retrieval, and source/page-level citation tracking.
- Integrated Tavily web search, BeautifulSoup scraping, Markdown report rendering, progress tracking, PDF upload, copy/download features, and API error handling in a responsive React UI.

•Data2Model | Distributed deep learning platform with automated tuning and monitoring

- Spearheaded a horizontally scalable, distributed Machine Learning pipeline using FastAPI, Celery, and Redis (Message Broker); streamlined Deep Learning hyperparameter tuning, reducing manual modeling overhead significantly.
- Formulated a fault-tolerant multi-node worker cluster to bypass strict 512MB RAM cloud constraints; implemented PyTorch memory throttling and parallel execution, effectively reducing training time by up to 50%.
- Incorporated the Google Gemini API to build an adaptive LLM-powered constraint engine; generated dataset specific neural network boundaries preventing OOM crashes, and synthesized real-time executive insights.
- Programmed a reactive React frontend synchronized with MongoDB Atlas for real-time state tracking; engineered robust external API fallbacks and cron-job keep-alives to ensure 99.9% system uptime on zero-cost instances.

•JobPortal | AI-Driven Recruitment & Resume Analysis Platform *Live Demo*

- Tech Stack: HTML, TailwindCSS, JavaScript, ReactJS (Vite), Node.js, Express.js, MongoDB
- Built a full-stack MERN job portal with Gemini AI-powered resume parsing and candidate skill scoring.
- Implemented JWT authentication with role-based access control for applicants, recruiters, and admins.
- Integrated Gemini API to auto-generate professional job descriptions from recruiter inputs.
- Used Cloudinary and Redux Toolkit for secure file uploads and efficient state management.

Technical Skills and Interests

- **Programming Languages:** C++, Python, JavaScript
- **AI/ML:** PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, LangChain, RAG, AI Agents, Vector Databases
- **Web & Databases:** React.js, Node.js, Express.js, MongoDB, MySQL
- **Core CS:** DSA, OOPs, Operating Systems, DBMS, Deep Learning, Computer Networks
- **Developer Tools:** Git, GitHub, VS Code, Google Colab

Achievements

- **AlgoUniversity Graph Contest** — Secured Institute Rank 6 *2025*
- **Google BigCode** — Qualified for Round 2; ranked among the top 1,500 participants. *2025*
- **Leetcode Biweekly Contest 152** — Secured Rank 327 among 30000+ participants *2025*
- **Guardian on Leetcode** — Reached the rating of 2169 *2026*
- **Specialist on Codeforces** — Reached the rating of 1446 *2026*

Coding Profiles

LeetCode: sumitabh2005 (Max: 2169) **Codeforces: Sumitabh** (Max: 1446) **CodeChef: sumitabh** (Max: 1618)

Positions of Responsibility

- **Member, Jigyasa Club** *Aug 2023*